SOLVED PRACTICE WORKBOOK

Defence R&D: DRDO and Missile Systems - Solved Practice Workbook

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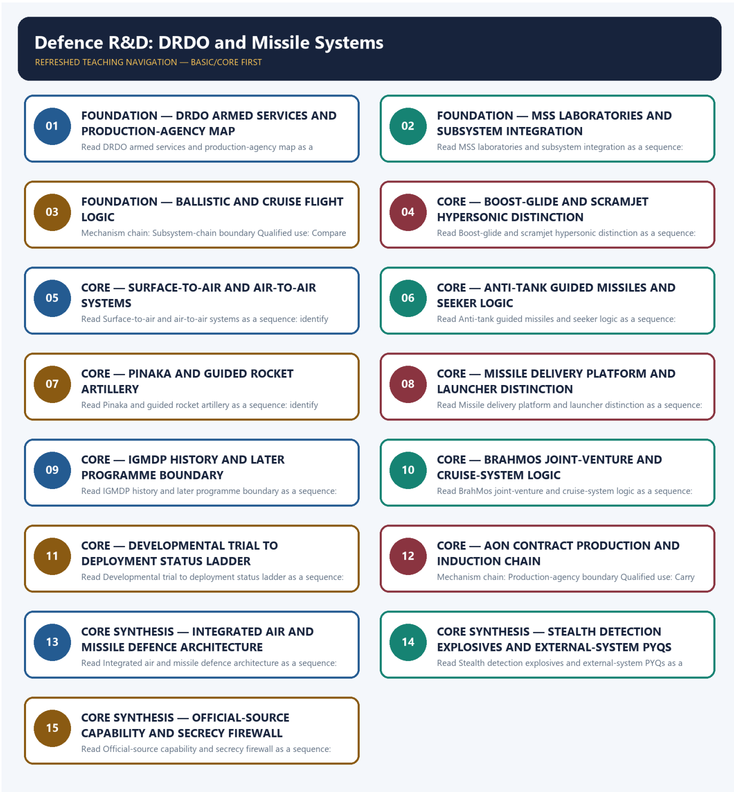

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Defence R&D: DRDO and Missile Systems - Solved Practice Workbook

BASIC MCQS / REMEDIATION

Q1. Which statement correctly identifies DRDO-developer boundary?

A. DRDO is the Ministry of Defence research-and-development organisation: it designs, develops, integrates and tests systems and facilitates production and induction; the armed services are users and production agencies manufacture at scale. B. A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an integrated operational weapon. C. The Army, Navy and Air Force articulate operational requirements, participate in user evaluation and decide service acceptance; a DRDO developmental result cannot be rewritten as service induction. D. The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres.

Answer: A. Explanation: DRDO is the Ministry of Defence research-and-development organisation: it designs, develops, integrates and tests systems and facilitates production and induction; the armed services are users and production agencies manufacture at scale. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q2. Which option preserves the technical boundary of DRDO-developer boundary?

A. A ballistic missile receives powered boost and then follows a largely ballistic trajectory, whereas a cruise missile sustains guided atmospheric flight using an air-breathing engine; range alone does not define either class. B. DRDO is the Ministry of Defence research-and-development organisation: it designs, develops, integrates and tests systems and facilitates production and induction; the armed services are users and production agencies manufacture at scale. C. The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres. D. A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an integrated operational weapon.

Answer: B. Explanation: DRDO is the Ministry of Defence research-and-development organisation: it designs, develops, integrates and tests systems and facilitates production and induction; the armed services are users and production agencies manufacture at scale. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q3. Which statement uses DRDO-developer boundary without changing its institution, unit or status?

A. Hypersonic describes speed rather than one missile class: a boost-glide vehicle is rocket-boosted and manoeuvres while gliding, while a hypersonic cruise system uses sustained air-breathing propulsion such as a scramjet. B. A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an integrated operational weapon. C. DRDO is the Ministry of Defence research-and-development organisation: it designs, develops, integrates and tests systems and facilitates production and induction; the armed services are users and production agencies manufacture at scale. D. A ballistic missile receives powered boost and then follows a largely ballistic trajectory, whereas a cruise missile sustains guided atmospheric flight using an air-breathing engine; range alone does not define either class.

Answer: C. Explanation: DRDO is the Ministry of Defence research-and-development organisation: it designs, develops, integrates and tests systems and facilitates production and induction; the armed services are users and production agencies manufacture at scale. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q4. Which option avoids the standard UPSC close-option trap about DRDO-developer boundary?

A. Hypersonic describes speed rather than one missile class: a boost-glide vehicle is rocket-boosted and manoeuvres while gliding, while a hypersonic cruise system uses sustained air-breathing propulsion such as a scramjet. B. A ballistic missile receives powered boost and then follows a largely ballistic trajectory, whereas a cruise missile sustains guided atmospheric flight using an air-breathing engine; range alone does not define either class. C. A surface-to-air missile is launched from the surface against aerial threats, while an air-to-air missile is launched from an aircraft; Akash and Astra therefore belong to different operational and launch-platform categories. D. DRDO is the Ministry of Defence research-and-development organisation: it designs, develops, integrates and tests systems and facilitates production and induction; the armed services are users and production agencies manufacture at scale.

Answer: D. Explanation: DRDO is the Ministry of Defence research-and-development organisation: it designs, develops, integrates and tests systems and facilitates production and induction; the armed services are users and production agencies manufacture at scale. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q5. Which statement correctly identifies Requirement-user boundary?

A. The Army, Navy and Air Force articulate operational requirements, participate in user evaluation and decide service acceptance; a DRDO developmental result cannot be rewritten as service induction. B. The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres. C. A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an integrated operational weapon. D. A ballistic missile receives powered boost and then follows a largely ballistic trajectory, whereas a cruise missile sustains guided atmospheric flight using an air-breathing engine; range alone does not define either class.

Answer: A. Explanation: The Army, Navy and Air Force articulate operational requirements, participate in user evaluation and decide service acceptance; a DRDO developmental result cannot be rewritten as service induction. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q6. Which option preserves the technical boundary of Requirement-user boundary?

A. A ballistic missile receives powered boost and then follows a largely ballistic trajectory, whereas a cruise missile sustains guided atmospheric flight using an air-breathing engine; range alone does not define either class. B. The Army, Navy and Air Force articulate operational requirements, participate in user evaluation and decide service acceptance; a DRDO developmental result cannot be rewritten as service induction. C. Hypersonic describes speed rather than one missile class: a boost-glide vehicle is rocket-boosted and manoeuvres while gliding, while a hypersonic cruise system uses sustained air-breathing propulsion such as a scramjet. D. A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an integrated operational weapon.

Answer: B. Explanation: The Army, Navy and Air Force articulate operational requirements, participate in user evaluation and decide service acceptance; a DRDO developmental result cannot be rewritten as service induction. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q7. Which statement uses Requirement-user boundary without changing its institution, unit or status?

A. Hypersonic describes speed rather than one missile class: a boost-glide vehicle is rocket-boosted and manoeuvres while gliding, while a hypersonic cruise system uses sustained air-breathing propulsion such as a scramjet. B. A ballistic missile receives powered boost and then follows a largely ballistic trajectory, whereas a cruise missile sustains guided atmospheric flight using an air-breathing engine; range alone does not define either class. C. The Army, Navy and Air Force articulate operational requirements, participate in user evaluation and decide service acceptance; a DRDO developmental result cannot be rewritten as service induction. D. A surface-to-air missile is launched from the surface against aerial threats, while an air-to-air missile is launched from an aircraft; Akash and Astra therefore belong to different operational and launch-platform categories.

Answer: C. Explanation: The Army, Navy and Air Force articulate operational requirements, participate in user evaluation and decide service acceptance; a DRDO developmental result cannot be rewritten as service induction. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q8. Which option avoids the standard UPSC close-option trap about Requirement-user boundary?

A. Hypersonic describes speed rather than one missile class: a boost-glide vehicle is rocket-boosted and manoeuvres while gliding, while a hypersonic cruise system uses sustained air-breathing propulsion such as a scramjet. B. A surface-to-air missile is launched from the surface against aerial threats, while an air-to-air missile is launched from an aircraft; Akash and Astra therefore belong to different operational and launch-platform categories. C. An anti-tank guided missile is designed for armoured targets and uses a guidance and seeker logic distinct from strategic ballistic, cruise and air-defence systems; the Nag family belongs to this role. D. The Army, Navy and Air Force articulate operational requirements, participate in user evaluation and decide service acceptance; a DRDO developmental result cannot be rewritten as service induction.

Answer: D. Explanation: The Army, Navy and Air Force articulate operational requirements, participate in user evaluation and decide service acceptance; a DRDO developmental result cannot be rewritten as service induction. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q9. Which statement correctly identifies MSS-cluster boundary?

A. The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres. B. A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an integrated operational weapon. C. A ballistic missile receives powered boost and then follows a largely ballistic trajectory, whereas a cruise missile sustains guided atmospheric flight using an air-breathing engine; range alone does not define either class. D. Hypersonic describes speed rather than one missile class: a boost-glide vehicle is rocket-boosted and manoeuvres while gliding, while a hypersonic cruise system uses sustained air-breathing propulsion such as a scramjet.

Answer: A. Explanation: The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q10. Which option preserves the technical boundary of MSS-cluster boundary?

A. A ballistic missile receives powered boost and then follows a largely ballistic trajectory, whereas a cruise missile sustains guided atmospheric flight using an air-breathing engine; range alone does not define either class. B. The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres. C. A surface-to-air missile is launched from the surface against aerial threats, while an air-to-air missile is launched from an aircraft; Akash and Astra therefore belong to different operational and launch-platform categories. D. Hypersonic describes speed rather than one missile class: a boost-glide vehicle is rocket-boosted and manoeuvres while gliding, while a hypersonic cruise system uses sustained air-breathing propulsion such as a scramjet.

Answer: B. Explanation: The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q11. Which statement uses MSS-cluster boundary without changing its institution, unit or status?

A. Hypersonic describes speed rather than one missile class: a boost-glide vehicle is rocket-boosted and manoeuvres while gliding, while a hypersonic cruise system uses sustained air-breathing propulsion such as a scramjet. B. A surface-to-air missile is launched from the surface against aerial threats, while an air-to-air missile is launched from an aircraft; Akash and Astra therefore belong to different operational and launch-platform categories. C. The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres. D. An anti-tank guided missile is designed for armoured targets and uses a guidance and seeker logic distinct from strategic ballistic, cruise and air-defence systems; the Nag family belongs to this role.

Answer: C. Explanation: The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q12. Which option avoids the standard UPSC close-option trap about MSS-cluster boundary?

A. A surface-to-air missile is launched from the surface against aerial threats, while an air-to-air missile is launched from an aircraft; Akash and Astra therefore belong to different operational and launch-platform categories. B. Pinaka is a multi-barrel rocket and guided-rocket artillery family for land fires; sharing rocket propulsion does not turn it into a strategic ballistic missile, cruise missile or space launch vehicle. C. An anti-tank guided missile is designed for armoured targets and uses a guidance and seeker logic distinct from strategic ballistic, cruise and air-defence systems; the Nag family belongs to this role. D. The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres.

Answer: D. Explanation: The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q13. Which statement correctly identifies Subsystem-chain boundary?

A. A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an integrated operational weapon. B. A surface-to-air missile is launched from the surface against aerial threats, while an air-to-air missile is launched from an aircraft; Akash and Astra therefore belong to different operational and launch-platform categories. C. Hypersonic describes speed rather than one missile class: a boost-glide vehicle is rocket-boosted and manoeuvres while gliding, while a hypersonic cruise system uses sustained air-breathing propulsion such as a scramjet. D. A ballistic missile receives powered boost and then follows a largely ballistic trajectory, whereas a cruise missile sustains guided atmospheric flight using an air-breathing engine; range alone does not define either class.

Answer: A. Explanation: A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an integrated operational weapon. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q14. Which option preserves the technical boundary of Subsystem-chain boundary?

A. An anti-tank guided missile is designed for armoured targets and uses a guidance and seeker logic distinct from strategic ballistic, cruise and air-defence systems; the Nag family belongs to this role. B. A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an integrated operational weapon. C. Hypersonic describes speed rather than one missile class: a boost-glide vehicle is rocket-boosted and manoeuvres while gliding, while a hypersonic cruise system uses sustained air-breathing propulsion such as a scramjet. D. A surface-to-air missile is launched from the surface against aerial threats, while an air-to-air missile is launched from an aircraft; Akash and Astra therefore belong to different operational and launch-platform categories.

Answer: B. Explanation: A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an integrated operational weapon. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q15. Which statement uses Subsystem-chain boundary without changing its institution, unit or status?

A. Pinaka is a multi-barrel rocket and guided-rocket artillery family for land fires; sharing rocket propulsion does not turn it into a strategic ballistic missile, cruise missile or space launch vehicle. B. A surface-to-air missile is launched from the surface against aerial threats, while an air-to-air missile is launched from an aircraft; Akash and Astra therefore belong to different operational and launch-platform categories. C. A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an integrated operational weapon. D. An anti-tank guided missile is designed for armoured targets and uses a guidance and seeker logic distinct from strategic ballistic, cruise and air-defence systems; the Nag family belongs to this role.

Answer: C. Explanation: A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an integrated operational weapon. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q16. Which option avoids the standard UPSC close-option trap about Subsystem-chain boundary?

A. Pinaka is a multi-barrel rocket and guided-rocket artillery family for land fires; sharing rocket propulsion does not turn it into a strategic ballistic missile, cruise missile or space launch vehicle. B. An anti-tank guided missile is designed for armoured targets and uses a guidance and seeker logic distinct from strategic ballistic, cruise and air-defence systems; the Nag family belongs to this role. C. A missile is the guided weapon, while an aircraft, ship, submarine, vehicle or launcher is the delivery platform; one missile may have multiple launch configurations without becoming the platform itself. D. A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an integrated operational weapon.

Answer: D. Explanation: A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an integrated operational weapon. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q17. Which statement correctly identifies Ballistic-cruise boundary?

A. A ballistic missile receives powered boost and then follows a largely ballistic trajectory, whereas a cruise missile sustains guided atmospheric flight using an air-breathing engine; range alone does not define either class. B. Hypersonic describes speed rather than one missile class: a boost-glide vehicle is rocket-boosted and manoeuvres while gliding, while a hypersonic cruise system uses sustained air-breathing propulsion such as a scramjet. C. A surface-to-air missile is launched from the surface against aerial threats, while an air-to-air missile is launched from an aircraft; Akash and Astra therefore belong to different operational and launch-platform categories. D. An anti-tank guided missile is designed for armoured targets and uses a guidance and seeker logic distinct from strategic ballistic, cruise and air-defence systems; the Nag family belongs to this role.

Answer: A. Explanation: A ballistic missile receives powered boost and then follows a largely ballistic trajectory, whereas a cruise missile sustains guided atmospheric flight using an air-breathing engine; range alone does not define either class. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q18. Which option preserves the technical boundary of Ballistic-cruise boundary?

A. An anti-tank guided missile is designed for armoured targets and uses a guidance and seeker logic distinct from strategic ballistic, cruise and air-defence systems; the Nag family belongs to this role. B. A ballistic missile receives powered boost and then follows a largely ballistic trajectory, whereas a cruise missile sustains guided atmospheric flight using an air-breathing engine; range alone does not define either class. C. A surface-to-air missile is launched from the surface against aerial threats, while an air-to-air missile is launched from an aircraft; Akash and Astra therefore belong to different operational and launch-platform categories. D. Pinaka is a multi-barrel rocket and guided-rocket artillery family for land fires; sharing rocket propulsion does not turn it into a strategic ballistic missile, cruise missile or space launch vehicle.

Answer: B. Explanation: A ballistic missile receives powered boost and then follows a largely ballistic trajectory, whereas a cruise missile sustains guided atmospheric flight using an air-breathing engine; range alone does not define either class. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q19. Which statement uses Ballistic-cruise boundary without changing its institution, unit or status?

A. Pinaka is a multi-barrel rocket and guided-rocket artillery family for land fires; sharing rocket propulsion does not turn it into a strategic ballistic missile, cruise missile or space launch vehicle. B. A missile is the guided weapon, while an aircraft, ship, submarine, vehicle or launcher is the delivery platform; one missile may have multiple launch configurations without becoming the platform itself. C. A ballistic missile receives powered boost and then follows a largely ballistic trajectory, whereas a cruise missile sustains guided atmospheric flight using an air-breathing engine; range alone does not define either class. D. An anti-tank guided missile is designed for armoured targets and uses a guidance and seeker logic distinct from strategic ballistic, cruise and air-defence systems; the Nag family belongs to this role.

Answer: C. Explanation: A ballistic missile receives powered boost and then follows a largely ballistic trajectory, whereas a cruise missile sustains guided atmospheric flight using an air-breathing engine; range alone does not define either class. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q20. Which option avoids the standard UPSC close-option trap about Ballistic-cruise boundary?

A. A missile is the guided weapon, while an aircraft, ship, submarine, vehicle or launcher is the delivery platform; one missile may have multiple launch configurations without becoming the platform itself. B. Pinaka is a multi-barrel rocket and guided-rocket artillery family for land fires; sharing rocket propulsion does not turn it into a strategic ballistic missile, cruise missile or space launch vehicle. C. The Integrated Guided Missile Development Programme ran from 1983 to March 2012 and covered Prithvi, Trishul, Akash, Nag and the Agni technology demonstrator; later families are not automatically IGMDP products. D. A ballistic missile receives powered boost and then follows a largely ballistic trajectory, whereas a cruise missile sustains guided atmospheric flight using an air-breathing engine; range alone does not define either class.

Answer: D. Explanation: A ballistic missile receives powered boost and then follows a largely ballistic trajectory, whereas a cruise missile sustains guided atmospheric flight using an air-breathing engine; range alone does not define either class. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q21. Which statement correctly identifies Hypersonic-class boundary?

A. Hypersonic describes speed rather than one missile class: a boost-glide vehicle is rocket-boosted and manoeuvres while gliding, while a hypersonic cruise system uses sustained air-breathing propulsion such as a scramjet. B. An anti-tank guided missile is designed for armoured targets and uses a guidance and seeker logic distinct from strategic ballistic, cruise and air-defence systems; the Nag family belongs to this role. C. A surface-to-air missile is launched from the surface against aerial threats, while an air-to-air missile is launched from an aircraft; Akash and Astra therefore belong to different operational and launch-platform categories. D. Pinaka is a multi-barrel rocket and guided-rocket artillery family for land fires; sharing rocket propulsion does not turn it into a strategic ballistic missile, cruise missile or space launch vehicle.

Answer: A. Explanation: Hypersonic describes speed rather than one missile class: a boost-glide vehicle is rocket-boosted and manoeuvres while gliding, while a hypersonic cruise system uses sustained air-breathing propulsion such as a scramjet. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q22. Which option preserves the technical boundary of Hypersonic-class boundary?

A. Pinaka is a multi-barrel rocket and guided-rocket artillery family for land fires; sharing rocket propulsion does not turn it into a strategic ballistic missile, cruise missile or space launch vehicle. B. Hypersonic describes speed rather than one missile class: a boost-glide vehicle is rocket-boosted and manoeuvres while gliding, while a hypersonic cruise system uses sustained air-breathing propulsion such as a scramjet. C. A missile is the guided weapon, while an aircraft, ship, submarine, vehicle or launcher is the delivery platform; one missile may have multiple launch configurations without becoming the platform itself. D. An anti-tank guided missile is designed for armoured targets and uses a guidance and seeker logic distinct from strategic ballistic, cruise and air-defence systems; the Nag family belongs to this role.

Answer: B. Explanation: Hypersonic describes speed rather than one missile class: a boost-glide vehicle is rocket-boosted and manoeuvres while gliding, while a hypersonic cruise system uses sustained air-breathing propulsion such as a scramjet. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q23. Which statement uses Hypersonic-class boundary without changing its institution, unit or status?

A. Pinaka is a multi-barrel rocket and guided-rocket artillery family for land fires; sharing rocket propulsion does not turn it into a strategic ballistic missile, cruise missile or space launch vehicle. B. A missile is the guided weapon, while an aircraft, ship, submarine, vehicle or launcher is the delivery platform; one missile may have multiple launch configurations without becoming the platform itself. C. Hypersonic describes speed rather than one missile class: a boost-glide vehicle is rocket-boosted and manoeuvres while gliding, while a hypersonic cruise system uses sustained air-breathing propulsion such as a scramjet. D. The Integrated Guided Missile Development Programme ran from 1983 to March 2012 and covered Prithvi, Trishul, Akash, Nag and the Agni technology demonstrator; later families are not automatically IGMDP products.

Answer: C. Explanation: Hypersonic describes speed rather than one missile class: a boost-glide vehicle is rocket-boosted and manoeuvres while gliding, while a hypersonic cruise system uses sustained air-breathing propulsion such as a scramjet. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q24. Which option avoids the standard UPSC close-option trap about Hypersonic-class boundary?

A. The Integrated Guided Missile Development Programme ran from 1983 to March 2012 and covered Prithvi, Trishul, Akash, Nag and the Agni technology demonstrator; later families are not automatically IGMDP products. B. BrahMos is a supersonic cruise-missile system developed and produced through BrahMos Aerospace, the DRDO-NPOM joint venture; it is neither a purely solo domestic programme nor a ballistic missile. C. A missile is the guided weapon, while an aircraft, ship, submarine, vehicle or launcher is the delivery platform; one missile may have multiple launch configurations without becoming the platform itself. D. Hypersonic describes speed rather than one missile class: a boost-glide vehicle is rocket-boosted and manoeuvres while gliding, while a hypersonic cruise system uses sustained air-breathing propulsion such as a scramjet.

Answer: D. Explanation: Hypersonic describes speed rather than one missile class: a boost-glide vehicle is rocket-boosted and manoeuvres while gliding, while a hypersonic cruise system uses sustained air-breathing propulsion such as a scramjet. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q25. Which statement correctly identifies SAM-AAM boundary?

A. A surface-to-air missile is launched from the surface against aerial threats, while an air-to-air missile is launched from an aircraft; Akash and Astra therefore belong to different operational and launch-platform categories. B. Pinaka is a multi-barrel rocket and guided-rocket artillery family for land fires; sharing rocket propulsion does not turn it into a strategic ballistic missile, cruise missile or space launch vehicle. C. A missile is the guided weapon, while an aircraft, ship, submarine, vehicle or launcher is the delivery platform; one missile may have multiple launch configurations without becoming the platform itself. D. An anti-tank guided missile is designed for armoured targets and uses a guidance and seeker logic distinct from strategic ballistic, cruise and air-defence systems; the Nag family belongs to this role.

Answer: A. Explanation: A surface-to-air missile is launched from the surface against aerial threats, while an air-to-air missile is launched from an aircraft; Akash and Astra therefore belong to different operational and launch-platform categories. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q26. Which option preserves the technical boundary of SAM-AAM boundary?

A. The Integrated Guided Missile Development Programme ran from 1983 to March 2012 and covered Prithvi, Trishul, Akash, Nag and the Agni technology demonstrator; later families are not automatically IGMDP products. B. A surface-to-air missile is launched from the surface against aerial threats, while an air-to-air missile is launched from an aircraft; Akash and Astra therefore belong to different operational and launch-platform categories. C. Pinaka is a multi-barrel rocket and guided-rocket artillery family for land fires; sharing rocket propulsion does not turn it into a strategic ballistic missile, cruise missile or space launch vehicle. D. A missile is the guided weapon, while an aircraft, ship, submarine, vehicle or launcher is the delivery platform; one missile may have multiple launch configurations without becoming the platform itself.

Answer: B. Explanation: A surface-to-air missile is launched from the surface against aerial threats, while an air-to-air missile is launched from an aircraft; Akash and Astra therefore belong to different operational and launch-platform categories. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q27. Which statement uses SAM-AAM boundary without changing its institution, unit or status?

A. The Integrated Guided Missile Development Programme ran from 1983 to March 2012 and covered Prithvi, Trishul, Akash, Nag and the Agni technology demonstrator; later families are not automatically IGMDP products. B. BrahMos is a supersonic cruise-missile system developed and produced through BrahMos Aerospace, the DRDO-NPOM joint venture; it is neither a purely solo domestic programme nor a ballistic missile. C. A surface-to-air missile is launched from the surface against aerial threats, while an air-to-air missile is launched from an aircraft; Akash and Astra therefore belong to different operational and launch-platform categories. D. A missile is the guided weapon, while an aircraft, ship, submarine, vehicle or launcher is the delivery platform; one missile may have multiple launch configurations without becoming the platform itself.

Answer: C. Explanation: A surface-to-air missile is launched from the surface against aerial threats, while an air-to-air missile is launched from an aircraft; Akash and Astra therefore belong to different operational and launch-platform categories. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q28. Which option avoids the standard UPSC close-option trap about SAM-AAM boundary?

A. BrahMos is a supersonic cruise-missile system developed and produced through BrahMos Aerospace, the DRDO-NPOM joint venture; it is neither a purely solo domestic programme nor a ballistic missile. B. Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested. C. The Integrated Guided Missile Development Programme ran from 1983 to March 2012 and covered Prithvi, Trishul, Akash, Nag and the Agni technology demonstrator; later families are not automatically IGMDP products. D. A surface-to-air missile is launched from the surface against aerial threats, while an air-to-air missile is launched from an aircraft; Akash and Astra therefore belong to different operational and launch-platform categories.

Answer: D. Explanation: A surface-to-air missile is launched from the surface against aerial threats, while an air-to-air missile is launched from an aircraft; Akash and Astra therefore belong to different operational and launch-platform categories. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q29. Which statement correctly identifies ATGM boundary?

A. An anti-tank guided missile is designed for armoured targets and uses a guidance and seeker logic distinct from strategic ballistic, cruise and air-defence systems; the Nag family belongs to this role. B. Pinaka is a multi-barrel rocket and guided-rocket artillery family for land fires; sharing rocket propulsion does not turn it into a strategic ballistic missile, cruise missile or space launch vehicle. C. A missile is the guided weapon, while an aircraft, ship, submarine, vehicle or launcher is the delivery platform; one missile may have multiple launch configurations without becoming the platform itself. D. The Integrated Guided Missile Development Programme ran from 1983 to March 2012 and covered Prithvi, Trishul, Akash, Nag and the Agni technology demonstrator; later families are not automatically IGMDP products.

Answer: A. Explanation: An anti-tank guided missile is designed for armoured targets and uses a guidance and seeker logic distinct from strategic ballistic, cruise and air-defence systems; the Nag family belongs to this role. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q30. Which option preserves the technical boundary of ATGM boundary?

A. BrahMos is a supersonic cruise-missile system developed and produced through BrahMos Aerospace, the DRDO-NPOM joint venture; it is neither a purely solo domestic programme nor a ballistic missile. B. An anti-tank guided missile is designed for armoured targets and uses a guidance and seeker logic distinct from strategic ballistic, cruise and air-defence systems; the Nag family belongs to this role. C. The Integrated Guided Missile Development Programme ran from 1983 to March 2012 and covered Prithvi, Trishul, Akash, Nag and the Agni technology demonstrator; later families are not automatically IGMDP products. D. A missile is the guided weapon, while an aircraft, ship, submarine, vehicle or launcher is the delivery platform; one missile may have multiple launch configurations without becoming the platform itself.

Answer: B. Explanation: An anti-tank guided missile is designed for armoured targets and uses a guidance and seeker logic distinct from strategic ballistic, cruise and air-defence systems; the Nag family belongs to this role. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q31. Which statement uses ATGM boundary without changing its institution, unit or status?

A. BrahMos is a supersonic cruise-missile system developed and produced through BrahMos Aerospace, the DRDO-NPOM joint venture; it is neither a purely solo domestic programme nor a ballistic missile. B. Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested. C. An anti-tank guided missile is designed for armoured targets and uses a guidance and seeker logic distinct from strategic ballistic, cruise and air-defence systems; the Nag family belongs to this role. D. The Integrated Guided Missile Development Programme ran from 1983 to March 2012 and covered Prithvi, Trishul, Akash, Nag and the Agni technology demonstrator; later families are not automatically IGMDP products.

Answer: C. Explanation: An anti-tank guided missile is designed for armoured targets and uses a guidance and seeker logic distinct from strategic ballistic, cruise and air-defence systems; the Nag family belongs to this role. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q32. Which option avoids the standard UPSC close-option trap about ATGM boundary?

A. Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested. B. Defence Acquisition Council Acceptance of Necessity is in-principle approval of a requirement and category, not a signed contract, delivery, induction or operational deployment. C. BrahMos is a supersonic cruise-missile system developed and produced through BrahMos Aerospace, the DRDO-NPOM joint venture; it is neither a purely solo domestic programme nor a ballistic missile. D. An anti-tank guided missile is designed for armoured targets and uses a guidance and seeker logic distinct from strategic ballistic, cruise and air-defence systems; the Nag family belongs to this role.

Answer: D. Explanation: An anti-tank guided missile is designed for armoured targets and uses a guidance and seeker logic distinct from strategic ballistic, cruise and air-defence systems; the Nag family belongs to this role. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q33. Which statement correctly identifies Rocket-artillery boundary?

A. Pinaka is a multi-barrel rocket and guided-rocket artillery family for land fires; sharing rocket propulsion does not turn it into a strategic ballistic missile, cruise missile or space launch vehicle. B. A missile is the guided weapon, while an aircraft, ship, submarine, vehicle or launcher is the delivery platform; one missile may have multiple launch configurations without becoming the platform itself. C. BrahMos is a supersonic cruise-missile system developed and produced through BrahMos Aerospace, the DRDO-NPOM joint venture; it is neither a purely solo domestic programme nor a ballistic missile. D. The Integrated Guided Missile Development Programme ran from 1983 to March 2012 and covered Prithvi, Trishul, Akash, Nag and the Agni technology demonstrator; later families are not automatically IGMDP products.

Answer: A. Explanation: Pinaka is a multi-barrel rocket and guided-rocket artillery family for land fires; sharing rocket propulsion does not turn it into a strategic ballistic missile, cruise missile or space launch vehicle. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q34. Which option preserves the technical boundary of Rocket-artillery boundary?

A. BrahMos is a supersonic cruise-missile system developed and produced through BrahMos Aerospace, the DRDO-NPOM joint venture; it is neither a purely solo domestic programme nor a ballistic missile. B. Pinaka is a multi-barrel rocket and guided-rocket artillery family for land fires; sharing rocket propulsion does not turn it into a strategic ballistic missile, cruise missile or space launch vehicle. C. Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested. D. The Integrated Guided Missile Development Programme ran from 1983 to March 2012 and covered Prithvi, Trishul, Akash, Nag and the Agni technology demonstrator; later families are not automatically IGMDP products.

Answer: B. Explanation: Pinaka is a multi-barrel rocket and guided-rocket artillery family for land fires; sharing rocket propulsion does not turn it into a strategic ballistic missile, cruise missile or space launch vehicle. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q35. Which statement uses Rocket-artillery boundary without changing its institution, unit or status?

A. Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested. B. BrahMos is a supersonic cruise-missile system developed and produced through BrahMos Aerospace, the DRDO-NPOM joint venture; it is neither a purely solo domestic programme nor a ballistic missile. C. Pinaka is a multi-barrel rocket and guided-rocket artillery family for land fires; sharing rocket propulsion does not turn it into a strategic ballistic missile, cruise missile or space launch vehicle. D. Defence Acquisition Council Acceptance of Necessity is in-principle approval of a requirement and category, not a signed contract, delivery, induction or operational deployment.

Answer: C. Explanation: Pinaka is a multi-barrel rocket and guided-rocket artillery family for land fires; sharing rocket propulsion does not turn it into a strategic ballistic missile, cruise missile or space launch vehicle. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q36. Which option avoids the standard UPSC close-option trap about Rocket-artillery boundary?

A. DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system. B. Defence Acquisition Council Acceptance of Necessity is in-principle approval of a requirement and category, not a signed contract, delivery, induction or operational deployment. C. Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested. D. Pinaka is a multi-barrel rocket and guided-rocket artillery family for land fires; sharing rocket propulsion does not turn it into a strategic ballistic missile, cruise missile or space launch vehicle.

Answer: D. Explanation: Pinaka is a multi-barrel rocket and guided-rocket artillery family for land fires; sharing rocket propulsion does not turn it into a strategic ballistic missile, cruise missile or space launch vehicle. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q37. Which statement correctly identifies Missile-platform boundary?

A. A missile is the guided weapon, while an aircraft, ship, submarine, vehicle or launcher is the delivery platform; one missile may have multiple launch configurations without becoming the platform itself. B. Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested. C. BrahMos is a supersonic cruise-missile system developed and produced through BrahMos Aerospace, the DRDO-NPOM joint venture; it is neither a purely solo domestic programme nor a ballistic missile. D. The Integrated Guided Missile Development Programme ran from 1983 to March 2012 and covered Prithvi, Trishul, Akash, Nag and the Agni technology demonstrator; later families are not automatically IGMDP products.

Answer: A. Explanation: A missile is the guided weapon, while an aircraft, ship, submarine, vehicle or launcher is the delivery platform; one missile may have multiple launch configurations without becoming the platform itself. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q38. Which option preserves the technical boundary of Missile-platform boundary?

A. BrahMos is a supersonic cruise-missile system developed and produced through BrahMos Aerospace, the DRDO-NPOM joint venture; it is neither a purely solo domestic programme nor a ballistic missile. B. A missile is the guided weapon, while an aircraft, ship, submarine, vehicle or launcher is the delivery platform; one missile may have multiple launch configurations without becoming the platform itself. C. Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested. D. Defence Acquisition Council Acceptance of Necessity is in-principle approval of a requirement and category, not a signed contract, delivery, induction or operational deployment.

Answer: B. Explanation: A missile is the guided weapon, while an aircraft, ship, submarine, vehicle or launcher is the delivery platform; one missile may have multiple launch configurations without becoming the platform itself. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q39. Which statement uses Missile-platform boundary without changing its institution, unit or status?

A. Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested. B. Defence Acquisition Council Acceptance of Necessity is in-principle approval of a requirement and category, not a signed contract, delivery, induction or operational deployment. C. A missile is the guided weapon, while an aircraft, ship, submarine, vehicle or launcher is the delivery platform; one missile may have multiple launch configurations without becoming the platform itself. D. DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system.

Answer: C. Explanation: A missile is the guided weapon, while an aircraft, ship, submarine, vehicle or launcher is the delivery platform; one missile may have multiple launch configurations without becoming the platform itself. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q40. Which option avoids the standard UPSC close-option trap about Missile-platform boundary?

A. DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system. B. Air and missile defence is a sensor, command-and-control and interceptor architecture; an announced integrated mission or one interceptor does not prove a nationwide deployed shield. C. Defence Acquisition Council Acceptance of Necessity is in-principle approval of a requirement and category, not a signed contract, delivery, induction or operational deployment. D. A missile is the guided weapon, while an aircraft, ship, submarine, vehicle or launcher is the delivery platform; one missile may have multiple launch configurations without becoming the platform itself.

Answer: D. Explanation: A missile is the guided weapon, while an aircraft, ship, submarine, vehicle or launcher is the delivery platform; one missile may have multiple launch configurations without becoming the platform itself. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q41. Which statement correctly identifies IGMDP-history boundary?

A. The Integrated Guided Missile Development Programme ran from 1983 to March 2012 and covered Prithvi, Trishul, Akash, Nag and the Agni technology demonstrator; later families are not automatically IGMDP products. B. Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested. C. BrahMos is a supersonic cruise-missile system developed and produced through BrahMos Aerospace, the DRDO-NPOM joint venture; it is neither a purely solo domestic programme nor a ballistic missile. D. Defence Acquisition Council Acceptance of Necessity is in-principle approval of a requirement and category, not a signed contract, delivery, induction or operational deployment.

Answer: A. Explanation: The Integrated Guided Missile Development Programme ran from 1983 to March 2012 and covered Prithvi, Trishul, Akash, Nag and the Agni technology demonstrator; later families are not automatically IGMDP products. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q42. Which option preserves the technical boundary of IGMDP-history boundary?

A. DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system. B. The Integrated Guided Missile Development Programme ran from 1983 to March 2012 and covered Prithvi, Trishul, Akash, Nag and the Agni technology demonstrator; later families are not automatically IGMDP products. C. Defence Acquisition Council Acceptance of Necessity is in-principle approval of a requirement and category, not a signed contract, delivery, induction or operational deployment. D. Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested.

Answer: B. Explanation: The Integrated Guided Missile Development Programme ran from 1983 to March 2012 and covered Prithvi, Trishul, Akash, Nag and the Agni technology demonstrator; later families are not automatically IGMDP products. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q43. Which statement uses IGMDP-history boundary without changing its institution, unit or status?

A. Air and missile defence is a sensor, command-and-control and interceptor architecture; an announced integrated mission or one interceptor does not prove a nationwide deployed shield. B. DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system. C. The Integrated Guided Missile Development Programme ran from 1983 to March 2012 and covered Prithvi, Trishul, Akash, Nag and the Agni technology demonstrator; later families are not automatically IGMDP products. D. Defence Acquisition Council Acceptance of Necessity is in-principle approval of a requirement and category, not a signed contract, delivery, induction or operational deployment.

Answer: C. Explanation: The Integrated Guided Missile Development Programme ran from 1983 to March 2012 and covered Prithvi, Trishul, Akash, Nag and the Agni technology demonstrator; later families are not automatically IGMDP products. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q44. Which option avoids the standard UPSC close-option trap about IGMDP-history boundary?

A. DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system. B. Air and missile defence is a sensor, command-and-control and interceptor architecture; an announced integrated mission or one interceptor does not prove a nationwide deployed shield. C. Stealth seeks lower observability through shaping, materials and operating tactics, but it does not mean invisibility; detection depends on sensors, frequency, geometry, processing and the operational environment. D. The Integrated Guided Missile Development Programme ran from 1983 to March 2012 and covered Prithvi, Trishul, Akash, Nag and the Agni technology demonstrator; later families are not automatically IGMDP products.

Answer: D. Explanation: The Integrated Guided Missile Development Programme ran from 1983 to March 2012 and covered Prithvi, Trishul, Akash, Nag and the Agni technology demonstrator; later families are not automatically IGMDP products. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q45. Which statement correctly identifies BrahMos-JV boundary?

A. BrahMos is a supersonic cruise-missile system developed and produced through BrahMos Aerospace, the DRDO-NPOM joint venture; it is neither a purely solo domestic programme nor a ballistic missile. B. DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system. C. Defence Acquisition Council Acceptance of Necessity is in-principle approval of a requirement and category, not a signed contract, delivery, induction or operational deployment. D. Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested.

Answer: A. Explanation: BrahMos is a supersonic cruise-missile system developed and produced through BrahMos Aerospace, the DRDO-NPOM joint venture; it is neither a purely solo domestic programme nor a ballistic missile. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q46. Which option preserves the technical boundary of BrahMos-JV boundary?

A. Defence Acquisition Council Acceptance of Necessity is in-principle approval of a requirement and category, not a signed contract, delivery, induction or operational deployment. B. BrahMos is a supersonic cruise-missile system developed and produced through BrahMos Aerospace, the DRDO-NPOM joint venture; it is neither a purely solo domestic programme nor a ballistic missile. C. DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system. D. Air and missile defence is a sensor, command-and-control and interceptor architecture; an announced integrated mission or one interceptor does not prove a nationwide deployed shield.

Answer: B. Explanation: BrahMos is a supersonic cruise-missile system developed and produced through BrahMos Aerospace, the DRDO-NPOM joint venture; it is neither a purely solo domestic programme nor a ballistic missile. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q47. Which statement uses BrahMos-JV boundary without changing its institution, unit or status?

A. Air and missile defence is a sensor, command-and-control and interceptor architecture; an announced integrated mission or one interceptor does not prove a nationwide deployed shield. B. DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system. C. BrahMos is a supersonic cruise-missile system developed and produced through BrahMos Aerospace, the DRDO-NPOM joint venture; it is neither a purely solo domestic programme nor a ballistic missile. D. Stealth seeks lower observability through shaping, materials and operating tactics, but it does not mean invisibility; detection depends on sensors, frequency, geometry, processing and the operational environment.

Answer: C. Explanation: BrahMos is a supersonic cruise-missile system developed and produced through BrahMos Aerospace, the DRDO-NPOM joint venture; it is neither a purely solo domestic programme nor a ballistic missile. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q48. Which option avoids the standard UPSC close-option trap about BrahMos-JV boundary?

A. CL-20, HMX and LLM-105 are high-energy substances discussed as military explosives; a chemical-category fact supplies no handling, composition, warhead or deployment detail. B. Stealth seeks lower observability through shaping, materials and operating tactics, but it does not mean invisibility; detection depends on sensors, frequency, geometry, processing and the operational environment. C. Air and missile defence is a sensor, command-and-control and interceptor architecture; an announced integrated mission or one interceptor does not prove a nationwide deployed shield. D. BrahMos is a supersonic cruise-missile system developed and produced through BrahMos Aerospace, the DRDO-NPOM joint venture; it is neither a purely solo domestic programme nor a ballistic missile.

Answer: D. Explanation: BrahMos is a supersonic cruise-missile system developed and produced through BrahMos Aerospace, the DRDO-NPOM joint venture; it is neither a purely solo domestic programme nor a ballistic missile. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q49. Which statement correctly identifies Test-status ladder?

A. Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested. B. DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system. C. Defence Acquisition Council Acceptance of Necessity is in-principle approval of a requirement and category, not a signed contract, delivery, induction or operational deployment. D. Air and missile defence is a sensor, command-and-control and interceptor architecture; an announced integrated mission or one interceptor does not prove a nationwide deployed shield.

Answer: A. Explanation: Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and

configuration officially tested. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q50. Which option preserves the technical boundary of Test-status ladder?

A. Stealth seeks lower observability through shaping, materials and operating tactics, but it does not mean invisibility; detection depends on sensors, frequency, geometry, processing and the operational environment. B. Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested. C. DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system. D. Air and missile defence is a sensor, command-and-control and interceptor architecture; an announced integrated mission or one interceptor does not prove a nationwide deployed shield.

Answer: B. Explanation: Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q51. Which statement uses Test-status ladder without changing its institution, unit or status?

A. Stealth seeks lower observability through shaping, materials and operating tactics, but it does not mean invisibility; detection depends on sensors, frequency, geometry, processing and the operational environment. B. Air and missile defence is a sensor, command-and-control and interceptor architecture; an announced integrated mission or one interceptor does not prove a nationwide deployed shield. C. Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested. D. CL-20, HMX and LLM-105 are high-energy substances discussed as military explosives; a chemical-category fact supplies no handling, composition, warhead or deployment detail.

Answer: C. Explanation: Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q52. Which option avoids the standard UPSC close-option trap about Test-status ladder?

A. CL-20, HMX and LLM-105 are high-energy substances discussed as military explosives; a chemical-category fact supplies no handling, composition, warhead or deployment detail. B. THAAD is a United States terminal high-altitude anti-ballistic-missile defence system, while a Fractional Orbital Bombardment System is a partial-orbit strike concept; neither is an Indian missile programme. C. Stealth seeks lower observability through shaping, materials and operating tactics, but it does not mean invisibility; detection depends on sensors, frequency, geometry, processing and the operational environment. D. Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested.

Answer: D. Explanation: Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q53. Which statement correctly identifies AoN-contract boundary?

A. Defence Acquisition Council Acceptance of Necessity is in-principle approval of a requirement and category, not a signed contract, delivery, induction or operational deployment. B. DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system. C. Stealth seeks lower observability through shaping, materials and operating tactics, but it does not mean invisibility; detection depends on sensors, frequency, geometry, processing and the operational environment. D. Air and missile defence is a sensor, command-and-control and interceptor architecture; an announced integrated mission or one interceptor does not prove a nationwide deployed shield.

Answer: A. Explanation: Defence Acquisition Council Acceptance of Necessity is in-principle approval of a requirement and category, not a signed contract, delivery, induction or operational deployment. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q54. Which option preserves the technical boundary of AoN-contract boundary?

A. Stealth seeks lower observability through shaping, materials and operating tactics, but it does not mean invisibility; detection depends on sensors, frequency, geometry, processing and the operational environment. B. Defence Acquisition Council Acceptance of Necessity is in-principle approval of a requirement and category, not a signed contract, delivery, induction or operational deployment. C. CL-20, HMX and LLM-105 are high-energy substances discussed as military explosives; a chemical-category fact supplies no handling, composition, warhead or deployment detail. D. Air and missile defence is a sensor, command-and-control and interceptor architecture; an announced integrated mission or one interceptor does not prove a nationwide deployed shield.

Answer: B. Explanation: Defence Acquisition Council Acceptance of Necessity is in-principle approval of a requirement and category, not a signed contract, delivery, induction or operational deployment. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q55. Which statement uses AoN-contract boundary without changing its institution, unit or status?

A. THAAD is a United States terminal high-altitude anti-ballistic-missile defence system, while a Fractional Orbital Bombardment System is a partial-orbit strike concept; neither is an Indian missile programme. B. CL-20, HMX and LLM-105 are high-energy substances discussed as military explosives; a chemical-category fact supplies no handling, composition, warhead or deployment detail. C. Defence Acquisition Council Acceptance of Necessity is in-principle approval of a requirement and category, not a signed contract, delivery, induction or operational deployment. D. Stealth seeks lower observability through shaping, materials and operating tactics, but it does not mean invisibility; detection depends on sensors, frequency, geometry, processing and the operational environment.

Answer: C. Explanation: Defence Acquisition Council Acceptance of Necessity is in-principle approval of a requirement and category, not a signed contract, delivery, induction or operational deployment. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q56. Which option avoids the standard UPSC close-option trap about AoN-contract boundary?

A. CL-20, HMX and LLM-105 are high-energy substances discussed as military explosives; a chemical-category fact supplies no handling, composition, warhead or deployment detail. B. Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers. C. THAAD is a United States terminal high-altitude anti-ballistic-missile defence system, while a Fractional Orbital Bombardment System is a partial-orbit strike concept; neither is an Indian missile programme. D. Defence Acquisition Council Acceptance of Necessity is in-principle approval of a requirement and category, not a signed contract, delivery, induction or operational deployment.

Answer: D. Explanation: Defence Acquisition Council Acceptance of Necessity is in-principle approval of a requirement and category, not a signed contract, delivery, induction or operational deployment. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q57. Which statement correctly identifies Production-agency boundary?

A. DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system. B. CL-20, HMX and LLM-105 are high-energy substances discussed as military explosives; a chemical-category fact supplies no handling, composition, warhead or deployment detail. C. Air and missile defence is a sensor, command-and-control and interceptor architecture; an announced integrated mission or one interceptor does not prove a nationwide deployed shield. D. Stealth seeks lower observability through shaping, materials and operating tactics, but it does not mean invisibility; detection depends on sensors, frequency, geometry, processing and the operational environment.

Answer: A. Explanation: DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system. The other options change the scientific category, institutional owner, measurement, mission

rung or operating status.

Q58. Which option preserves the technical boundary of Production-agency boundary?

A. Stealth seeks lower observability through shaping, materials and operating tactics, but it does not mean invisibility; detection depends on sensors, frequency, geometry, processing and the operational environment. B. DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system. C. THAAD is a United States terminal high-altitude anti-ballistic-missile defence system, while a Fractional Orbital Bombardment System is a partial-orbit strike concept; neither is an Indian missile programme. D. CL-20, HMX and LLM-105 are high-energy substances discussed as military explosives; a chemical-category fact supplies no handling, composition, warhead or deployment detail.

Answer: B. Explanation: DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q59. Which statement uses Production-agency boundary without changing its institution, unit or status?

A. Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers. B. THAAD is a United States terminal high-altitude anti-ballistic-missile defence system, while a Fractional Orbital Bombardment System is a partial-orbit strike concept; neither is an Indian missile programme. C. DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system. D. CL-20, HMX and LLM-105 are high-energy substances discussed as military explosives; a chemical-category fact supplies no handling, composition, warhead or deployment detail.

Answer: C. Explanation: DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q60. Which option avoids the standard UPSC close-option trap about Production-agency boundary?

A. THAAD is a United States terminal high-altitude anti-ballistic-missile defence system, while a Fractional Orbital Bombardment System is a partial-orbit strike concept; neither is an Indian missile programme. B. DRDO is the Ministry of Defence research-and-development organisation: it designs, develops, integrates and tests systems and facilitates production and induction; the armed services are users and production agencies manufacture at scale. C. Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers. D. DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system.

Answer: D. Explanation: DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q61. Which statement correctly identifies Integrated-air-defence boundary?

A. Air and missile defence is a sensor, command-and-control and interceptor architecture; an announced integrated mission or one interceptor does not prove a nationwide deployed shield. B. THAAD is a United States terminal high-altitude anti-ballistic-missile defence system, while a Fractional Orbital Bombardment System is a partial-orbit strike concept; neither is an Indian missile programme. C. Stealth seeks lower observability through shaping, materials and operating tactics, but it does not mean invisibility; detection depends on sensors, frequency, geometry, processing and the operational environment. D. CL-20, HMX and LLM-105 are high-energy substances discussed as military explosives; a chemical-category fact supplies no handling, composition, warhead or deployment detail.

Answer: A. Explanation: Air and missile defence is a sensor, command-and-control and interceptor architecture; an announced integrated mission or one interceptor does not prove a nationwide deployed shield. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q62. Which option preserves the technical boundary of Integrated-air-defence boundary?

A. CL-20, HMX and LLM-105 are high-energy substances discussed as military explosives; a chemical-category fact supplies no handling, composition, warhead or deployment detail. B. Air and missile defence is a sensor, command-and-control and interceptor architecture; an announced integrated mission or one interceptor does not prove a nationwide deployed shield. C. Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers. D. THAAD is a United States terminal high-altitude anti-ballistic-missile defence system, while a Fractional Orbital Bombardment System is a partial-orbit strike concept; neither is an Indian missile programme.

Answer: B. Explanation: Air and missile defence is a sensor, command-and-control and interceptor architecture; an announced integrated mission or one interceptor does not prove a nationwide deployed shield. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q63. Which statement uses Integrated-air-defence boundary without changing its institution, unit or status?

A. THAAD is a United States terminal high-altitude anti-ballistic-missile defence system, while a Fractional Orbital Bombardment System is a partial-orbit strike concept; neither is an Indian missile programme. B. DRDO is the Ministry of Defence research-and-development organisation: it designs, develops, integrates and tests systems and facilitates production and induction; the armed services are users and production agencies manufacture at scale. C. Air and missile defence is a sensor, command-and-control and interceptor architecture; an announced integrated mission or one interceptor does not prove a nationwide deployed shield. D. Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers.

Answer: C. Explanation: Air and missile defence is a sensor, command-and-control and interceptor architecture; an announced integrated mission or one interceptor does not prove a nationwide deployed shield. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q64. Which option avoids the standard UPSC close-option trap about Integrated-air-defence boundary?

A. DRDO is the Ministry of Defence research-and-development organisation: it designs, develops, integrates and tests systems and facilitates production and induction; the armed services are users and production agencies manufacture at scale. B. The Army, Navy and Air Force articulate operational requirements, participate in user evaluation and decide service acceptance; a DRDO developmental result cannot be rewritten as service induction. C. Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers. D. Air and missile defence is a sensor, command-and-control and interceptor architecture; an announced integrated mission or one interceptor does not prove a nationwide deployed shield.

Answer: D. Explanation: Air and missile defence is a sensor, command-and-control and interceptor architecture; an announced integrated mission or one interceptor does not prove a nationwide deployed shield. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q65. Which statement correctly identifies Stealth-detection boundary?

A. Stealth seeks lower observability through shaping, materials and operating tactics, but it does not mean invisibility; detection depends on sensors, frequency, geometry, processing and the operational environment. B. THAAD is a United States terminal high-altitude anti-ballistic-missile defence system, while a Fractional Orbital Bombardment System is a partial-orbit strike concept; neither is an Indian missile programme. C. Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers. D. CL-20, HMX and LLM-105 are high-energy substances discussed as military explosives; a chemical-category fact supplies no handling, composition, warhead or deployment detail.

Answer: A. Explanation: Stealth seeks lower observability through shaping, materials and operating tactics, but it does not mean invisibility; detection depends on sensors, frequency, geometry, processing and the operational environment. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q66. Which option preserves the technical boundary of Stealth-detection boundary?

A. DRDO is the Ministry of Defence research-and-development organisation: it designs, develops, integrates and tests systems and facilitates production and induction; the armed services are users and production agencies manufacture at scale. B. Stealth seeks lower observability through shaping, materials and operating tactics, but it does not mean invisibility; detection depends on sensors, frequency, geometry, processing and the operational environment. C. Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers. D. THAAD is a United States terminal high-altitude anti-ballistic-missile defence system, while a Fractional Orbital Bombardment System is a partial-orbit strike concept; neither is an Indian missile programme.

Answer: B. Explanation: Stealth seeks lower observability through shaping, materials and operating tactics, but it does not mean invisibility; detection depends on sensors, frequency, geometry, processing and the operational environment. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q67. Which statement uses Stealth-detection boundary without changing its institution, unit or status?

A. The Army, Navy and Air Force articulate operational requirements, participate in user evaluation and decide service acceptance; a DRDO developmental result cannot be rewritten as service induction. B. Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers. C. Stealth seeks lower observability through shaping, materials and operating tactics, but it does not mean invisibility; detection depends on sensors, frequency, geometry, processing and the operational environment. D. DRDO is the Ministry of Defence research-and-development organisation: it designs, develops, integrates and tests systems and facilitates production and induction; the armed services are users and production agencies manufacture at scale.

Answer: C. Explanation: Stealth seeks lower observability through shaping, materials and operating tactics, but it does not mean invisibility; detection depends on sensors, frequency, geometry, processing and the operational environment. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q68. Which option avoids the standard UPSC close-option trap about Stealth-detection boundary?

A. The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres. B. The Army, Navy and Air Force articulate operational requirements, participate in user evaluation and decide service acceptance; a DRDO developmental result cannot be rewritten as service induction. C. DRDO is the Ministry of Defence research-and-development organisation: it designs, develops, integrates and tests systems and facilitates production and induction; the armed services are users and production agencies manufacture at scale. D. Stealth seeks lower observability through shaping, materials and operating tactics, but it does not mean invisibility; detection depends on sensors, frequency, geometry, processing and the operational environment.

Answer: D. Explanation: Stealth seeks lower observability through shaping, materials and operating tactics, but it does not mean invisibility; detection depends on sensors, frequency, geometry, processing and the operational environment. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q69. Which statement correctly identifies Explosives-category boundary?

A. CL-20, HMX and LLM-105 are high-energy substances discussed as military explosives; a chemical-category fact supplies no handling, composition, warhead or deployment detail. B. Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers. C. DRDO is the Ministry of Defence research-and-development organisation: it designs, develops, integrates and tests systems and facilitates production and induction; the armed services are users and production agencies manufacture at scale. D. THAAD is a United States terminal high-altitude anti-ballistic-missile defence system, while a Fractional Orbital Bombardment System is a partial-orbit strike concept; neither is an Indian missile programme.

Answer: A. Explanation: CL-20, HMX and LLM-105 are high-energy substances discussed as military explosives; a chemical-category fact supplies no handling, composition, warhead or deployment detail. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q70. Which option preserves the technical boundary of Explosives-category boundary?

A. Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers. B. CL-20, HMX and LLM-105 are high-energy substances discussed as military explosives; a chemical-category fact supplies no handling, composition, warhead or deployment detail. C. DRDO is the Ministry of Defence research-and-development organisation: it designs, develops, integrates and tests systems and facilitates production and induction; the armed services are users and production agencies manufacture at scale. D. The Army, Navy and Air Force articulate operational requirements, participate in user evaluation and decide service acceptance; a DRDO developmental result cannot be rewritten as service induction.

Answer: B. Explanation: CL-20, HMX and LLM-105 are high-energy substances discussed as military explosives; a chemical-category fact supplies no handling, composition, warhead or deployment detail. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q71. Which statement uses Explosives-category boundary without changing its institution, unit or status?

A. DRDO is the Ministry of Defence research-and-development organisation: it designs, develops, integrates and tests systems and facilitates production and induction; the armed services are users and production agencies manufacture at scale. B. The Army, Navy and Air Force articulate operational requirements, participate in user evaluation and decide service acceptance; a DRDO developmental result cannot be rewritten as service induction. C. CL-20, HMX and LLM-105 are high-energy substances discussed as military explosives; a chemical-category fact supplies no handling, composition, warhead or deployment detail. D. The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres.

Answer: C. Explanation: CL-20, HMX and LLM-105 are high-energy substances discussed as military explosives; a chemical-category fact supplies no handling, composition, warhead or deployment detail. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q72. Which option avoids the standard UPSC close-option trap about Explosives-category boundary?

A. The Army, Navy and Air Force articulate operational requirements, participate in user evaluation and decide service acceptance; a DRDO developmental result cannot be rewritten as service induction. B. The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres. C. A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an integrated operational weapon. D. CL-20, HMX and LLM-105 are high-energy substances discussed as military explosives; a chemical-category fact supplies no handling, composition, warhead or deployment detail.

Answer: D. Explanation: CL-20, HMX and LLM-105 are high-energy substances discussed as military explosives; a chemical-category fact supplies no handling, composition, warhead or deployment detail. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q73. Which statement correctly identifies THAAD-FOBS boundary?

A. THAAD is a United States terminal high-altitude anti-ballistic-missile defence system, while a Fractional Orbital Bombardment System is a partial-orbit strike concept; neither is an Indian missile programme. B. DRDO is the Ministry of Defence research-and-development organisation: it designs, develops, integrates and tests systems and facilitates production and induction; the armed services are users and production agencies manufacture at scale. C. The Army, Navy and Air Force articulate operational requirements, participate in user evaluation and decide service acceptance; a DRDO developmental result cannot be rewritten as service induction. D. Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers.

Answer: A. Explanation: THAAD is a United States terminal high-altitude anti-ballistic-missile defence system, while a Fractional Orbital Bombardment System is a partial-orbit strike concept; neither is an Indian missile programme. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q74. Which option preserves the technical boundary of THAAD-FOBS boundary?

A. DRDO is the Ministry of Defence research-and-development organisation: it designs, develops, integrates and tests systems and facilitates production and induction; the armed services are users and production agencies manufacture at scale. B. THAAD is a United States terminal high-altitude anti-ballistic-missile defence system, while a Fractional Orbital Bombardment System is a partial-orbit strike concept; neither is an Indian missile programme. C. The Army, Navy and Air Force articulate operational requirements, participate in user evaluation and decide service acceptance; a DRDO developmental result cannot be rewritten as service induction. D. The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres.

Answer: B. Explanation: THAAD is a United States terminal high-altitude anti-ballistic-missile defence system, while a Fractional Orbital Bombardment System is a partial-orbit strike concept; neither is an Indian missile programme. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q75. Which statement uses THAAD-FOBS boundary without changing its institution, unit or status?

A. A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an integrated operational weapon. B. The Army, Navy and Air Force articulate operational requirements, participate in user evaluation and decide service acceptance; a DRDO developmental result cannot be rewritten as service induction. C. THAAD is a United States terminal high-altitude anti-ballistic-missile defence system, while a Fractional Orbital Bombardment System is a partial-orbit strike concept; neither is an Indian missile programme. D. The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres.

Answer: C. Explanation: THAAD is a United States terminal high-altitude anti-ballistic-missile defence system, while a Fractional Orbital Bombardment System is a partial-orbit strike concept; neither is an Indian missile programme. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q76. Which option avoids the standard UPSC close-option trap about THAAD-FOBS boundary?

A. The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres. B. A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an integrated operational weapon. C. A ballistic missile receives powered boost and then follows a largely ballistic trajectory, whereas a cruise missile sustains guided atmospheric flight using an air-breathing engine; range alone does not define either class. D. THAAD is a United States terminal high-altitude anti-ballistic-missile defence system, while a Fractional Orbital Bombardment System is a partial-orbit strike concept; neither is an Indian missile programme.

Answer: D. Explanation: THAAD is a United States terminal high-altitude anti-ballistic-missile defence system, while a Fractional Orbital Bombardment System is a partial-orbit strike concept; neither is an Indian missile programme. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q77. Which statement correctly identifies Volatile-capability boundary?

A. Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers. B. The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres. C. DRDO is the Ministry of Defence research-and-development organisation: it designs, develops, integrates and tests systems and facilitates production and induction; the armed services are users and production agencies manufacture at scale. D. The Army, Navy and Air Force articulate operational requirements, participate in user evaluation and decide service acceptance; a DRDO developmental result cannot be rewritten as service induction.

Answer: A. Explanation: Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q78. Which option preserves the technical boundary of Volatile-capability boundary?

A. A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an integrated operational weapon. B. Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers. C. The Army, Navy and Air Force articulate operational requirements, participate in user evaluation and decide service acceptance; a DRDO developmental result cannot be rewritten as service induction. D. The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres.

Answer: B. Explanation: Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q79. Which statement uses Volatile-capability boundary without changing its institution, unit or status?

A. A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an integrated operational weapon. B. The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres. C. Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers. D. A ballistic missile receives powered boost and then follows a largely ballistic trajectory, whereas a cruise missile sustains guided atmospheric flight using an air-breathing engine; range alone does not define either class.

Answer: C. Explanation: Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q80. Which option avoids the standard UPSC close-option trap about Volatile-capability boundary?

A. A ballistic missile receives powered boost and then follows a largely ballistic trajectory, whereas a cruise missile sustains guided atmospheric flight using an air-breathing engine; range alone does not define either class. B. Hypersonic describes speed rather than one missile class: a boost-glide vehicle is rocket-boosted and manoeuvres while gliding, while a hypersonic cruise system uses sustained air-breathing propulsion such as a scramjet. C. A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an integrated operational weapon. D. Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers.

Answer: D. Explanation: Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

PYQS AND ANSWER PRACTICE

AUDITED TOPIC-SPECIFIC PYQ OWNERSHIP

Audited ledgers route the 2021 S-400 Mains demand, the 2023 ballistic-versus-cruise objective concept and later provisional defence-technology concepts here. No objective key, sensitive specification or deployment fact is invented.

PYQ DEMAND CARD 1 - 2021 GS-III

Demand: Explain the technical superiority of the S-400 air-defence system over other systems.

Status: Verified routed Mains demand; the answer uses sensor-command-interceptor architecture and does not invent classified Indian deployment detail.

Model solution: SAM-AAM boundary: A surface-to-air missile is launched from the surface against aerial threats, while an air-to-air missile is launched from an aircraft; Akash and Astra therefore belong to different operational and launch-platform categories. **Test-status ladder:** Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested. **Integrated-air-defence boundary:** Air and missile defence is a sensor, command-and-control and interceptor architecture; an announced integrated mission or one interceptor does not prove a nationwide deployed shield. **Volatile-capability boundary:** Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: The directive **answer** requires a direct position on “PYQ DEMAND CARD 1 - 2021 GS-III”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: SAM-AAM boundary: A surface-to-air missile is launched from the surface against aerial threats, while an air-to-air missile is launched from an aircraft; Akash and Astra therefore belong to different operational and launch-platform categories. **Test-status ladder:** Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested. **Integrated-air-defence boundary:** Air and missile defence is a sensor, command-and-control and interceptor architecture; an announced integrated mission or one interceptor does not prove a nationwide deployed shield. **Volatile-capability boundary:** Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim and named evidence:** Demand: Explain the technical superiority of the S-400 air-defence system over other systems. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Status: Verified routed Mains demand; the answer uses sensor-command-interceptor architecture and does not invent classified Indian deployment detail. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: SAM-AAM boundary: A surface-to-air missile is launched from the surface against aerial threats, while an air-to-air missile is launched from an aircraft; Akash and Astra therefore belong to different operational and launch-platform categories. **Test-status ladder:** Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested. **Integrated-air-defence boundary:** Air and missile defence is a sensor, command-and-control and interceptor architecture; an announced integrated mission or one interceptor does not prove a nationwide deployed shield. **Volatile-capability boundary:** Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “PYQ DEMAND CARD 1 - 2021 GS-III”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

PYQ DEMAND CARD 2 - 2023 Prelims GS-I

Demand: Distinguish Agni ballistic missiles from BrahMos cruise missiles.

Status: Verified routed objective concept; the local official answer key was unavailable, so no answer letter is asserted.

Model solution: Ballistic-cruise boundary: A ballistic missile receives powered boost and then follows a largely ballistic trajectory, whereas a cruise missile sustains guided atmospheric flight using an air-breathing engine; range alone does not define either class. **BrahMos-JV boundary:** BrahMos is a supersonic cruise-missile system developed and produced through BrahMos Aerospace, the DRDO-NPOM joint venture; it is neither a purely solo domestic programme nor a ballistic missile. **Volatile-capability boundary:** Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: Treat “PYQ DEMAND CARD 2 - 2023 Prelims GS-I” as a mechanism, category, institution, unit, date and status problem.

Detailed examiner-grade model answer:

Introduction and thesis: Ballistic-cruise boundary: A ballistic missile receives powered boost and then follows a largely ballistic trajectory, whereas a cruise missile sustains guided atmospheric flight using an air-breathing engine; range alone does not define either class. **BrahMos-JV boundary:** BrahMos is a supersonic cruise-missile system developed and produced through BrahMos Aerospace, the DRDO-NPOM joint venture; it is neither a purely solo domestic programme nor a ballistic missile. **Volatile-capability boundary:** Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim and named evidence:** Demand: Distinguish Agni ballistic missiles from BrahMos cruise missiles. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Status: Verified routed objective concept; the local official answer key was unavailable, so no answer letter is asserted. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Ballistic-cruise boundary: A ballistic missile receives powered boost and then follows a largely ballistic trajectory, whereas a cruise missile sustains guided atmospheric flight using an air-breathing engine; range alone does not define either class. **BrahMos-JV boundary:** BrahMos is a supersonic cruise-missile system developed and produced through BrahMos Aerospace, the DRDO-NPOM joint venture; it is neither a purely solo domestic programme nor a ballistic missile. **Volatile-capability boundary:** Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: Identify the object and actor; trace the technical mechanism; fix the measurement and platform; test each statement against source date, status rung and closest exception.

Why this earns marks: It prevents organisation, platform, process, application, regulatory role, target and observed outcome from being conflated.

How to improve this answer: For “PYQ DEMAND CARD 2 - 2023 Prelims GS-I”, explain why the nearest distractor fails on mechanism, platform, unit, institution, legal status, maturity or date.

PYQ DEMAND CARD 3 - 2026 Prelims GS-I

Demand: Assess stealth, radar cross-section and detection statements.

Status: Provisional routed concept; no option letter is recorded or inferred.

Model solution: Stealth-detection boundary: Stealth seeks lower observability through shaping, materials and operating tactics, but it does not mean invisibility; detection depends on sensors, frequency, geometry, processing and the operational environment. **Volatile-capability boundary:** Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: Treat "PYQ DEMAND CARD 3 - 2026 Prelims GS-I" as a mechanism, category, institution, unit, date and status problem.

Detailed examiner-grade model answer:

Introduction and thesis: Stealth-detection boundary: Stealth seeks lower observability through shaping, materials and operating tactics, but it does not mean invisibility; detection depends on sensors, frequency, geometry, processing and the operational environment. **Volatile-capability boundary:** Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim and named evidence:** Demand: Assess stealth, radar cross-section and detection statements. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Status: Provisional routed concept; no option letter is recorded or inferred. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Stealth-detection boundary: Stealth seeks lower observability through shaping, materials and operating tactics, but it does not mean invisibility; detection depends on sensors, frequency, geometry, processing and the operational environment. **Volatile-capability boundary:** Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: Identify the object and actor; trace the technical mechanism; fix the measurement and platform; test each statement against source date, status rung and closest exception.

Why this earns marks: It prevents organisation, platform, process, application, regulatory role, target and observed outcome from being conflated.

How to improve this answer: For "PYQ DEMAND CARD 3 - 2026 Prelims GS-I", explain why the nearest distractor fails on mechanism, platform, unit, institution, legal status, maturity or date.

ORIGINAL MAINS 1 - 10 MARKS

Question: Distinguish DRDO's role from the roles of the armed services and production agencies. Answer in about 150 words.

Model thesis: Claim: DRDO-developer boundary. **Named evidence/example:** DRDO is the Ministry of Defence research-and-development organisation: it designs, develops, integrates and tests systems and facilitates production and induction; the armed services are users and production agencies manufacture at scale. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Requirement-user boundary. **Named evidence/example:** The Army, Navy and Air Force articulate operational requirements, participate in user evaluation and decide service acceptance; a DRDO developmental result cannot be rewritten as service induction. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** MSS-cluster boundary. **Named evidence/example:** The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Production-agency boundary. **Named evidence/example:** DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- DRDO is the Ministry of Defence research-and-development organisation: it designs, develops, integrates and tests systems and facilitates production and induction; the armed services are users and production agencies manufacture at scale.
- The Army, Navy and Air Force articulate operational requirements, participate in user evaluation and decide service acceptance; a DRDO developmental result cannot be rewritten as service induction.
- The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres.
- DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system.

Qualified conclusion: Claim: DRDO-developer boundary. **Named evidence/example:** DRDO is the Ministry of Defence research-and-development organisation: it designs, develops, integrates and tests systems and facilitates production and induction; the armed services are users and production agencies manufacture at scale. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Requirement-user boundary. **Named evidence/example:** The Army, Navy and Air Force articulate operational requirements, participate in user evaluation and decide service acceptance; a DRDO developmental result cannot be rewritten as service induction. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** MSS-cluster boundary. **Named evidence/example:** The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Production-agency boundary. **Named evidence/example:** DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **answer** requires a direct position on “Distinguish DRDO's role from the roles of the armed services and production agencies. Answer…”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: DRDO-developer boundary. **Named evidence/example:** DRDO is the Ministry of Defence research-and-development organisation: it designs, develops, integrates and tests systems and facilitates production and induction; the armed services are users and production agencies manufacture at scale. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Requirement-user boundary. **Named evidence/example:** The Army, Navy and Air Force articulate operational requirements, participate in user evaluation and decide service acceptance; a DRDO developmental result cannot be rewritten as service induction. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** MSS-cluster boundary. **Named evidence/example:** The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Production-agency boundary. **Named evidence/example:** DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** DRDO is the Ministry of Defence research-and-development organisation: it designs, develops, integrates and tests systems and facilitates production and induction; the armed services are users and production agencies manufacture at scale. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** The Army, Navy and Air Force articulate operational requirements, participate in user evaluation and decide service acceptance; a DRDO developmental result cannot be rewritten as service induction. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: DRDO-developer boundary. **Named evidence/example:** DRDO is the Ministry of Defence research-and-development organisation: it designs, develops, integrates and tests systems and facilitates production and induction; the armed services are users and production agencies manufacture at scale. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Requirement-user boundary. **Named evidence/example:** The Army, Navy and Air Force articulate operational requirements, participate in user evaluation and decide service acceptance; a DRDO developmental result cannot be rewritten as service induction. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** MSS-cluster boundary. **Named evidence/example:** The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Production-agency boundary. **Named evidence/example:** DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “Distinguish DRDO's role from the roles of the armed services and production agencies. Answer…”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 2 - 10 MARKS

Question: Differentiate ballistic, cruise and guided rocket-artillery systems. Answer in about 150 words.

Model thesis: Claim: Ballistic-cruise boundary. **Named evidence/example:** A ballistic missile receives powered boost and then follows a largely ballistic trajectory, whereas a cruise missile sustains guided atmospheric flight using an air-breathing engine; range alone does not define either class. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Hypersonic-class boundary. **Named evidence/example:** Hypersonic describes speed rather than one missile class: a boost-glide vehicle is rocket-boosted and manoeuvres while gliding, while a hypersonic cruise system uses sustained air-breathing propulsion such as a scramjet. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Rocket-artillery boundary. **Named evidence/example:** Pinaka is a multi-barrel rocket and guided-rocket artillery family for land fires; sharing rocket propulsion does not turn it into a strategic ballistic missile, cruise missile or space launch vehicle. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Missile-platform boundary. **Named evidence/example:** A missile is the guided weapon, while an aircraft, ship, submarine, vehicle or launcher is the delivery platform; one missile may have multiple launch configurations without becoming the platform itself. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- A ballistic missile receives powered boost and then follows a largely ballistic trajectory, whereas a cruise missile sustains guided atmospheric flight using an air-breathing engine; range alone does not define either class.
- Hypersonic describes speed rather than one missile class: a boost-glide vehicle is rocket-boosted and manoeuvres while gliding, while a hypersonic cruise system uses sustained air-breathing propulsion such as a scramjet.
- Pinaka is a multi-barrel rocket and guided-rocket artillery family for land fires; sharing rocket propulsion does not turn it into a strategic ballistic missile, cruise missile or space launch vehicle.
- A missile is the guided weapon, while an aircraft, ship, submarine, vehicle or launcher is the delivery platform; one missile may have multiple launch configurations without becoming the platform itself.

Qualified conclusion: Claim: Ballistic-cruise boundary. **Named evidence/example:** A ballistic missile receives powered boost and then follows a largely ballistic trajectory, whereas a cruise missile sustains guided atmospheric flight using an air-breathing engine; range alone does not define either class. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Hypersonic-class boundary. **Named evidence/example:** Hypersonic describes speed rather than one missile class: a boost-glide vehicle is rocket-boosted and manoeuvres while gliding, while a hypersonic cruise system uses sustained air-breathing propulsion such as a scramjet. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Rocket-artillery boundary. **Named evidence/example:** Pinaka is a multi-barrel rocket and guided-rocket artillery family for land fires; sharing rocket propulsion does not turn it into a strategic ballistic missile, cruise missile or space launch vehicle. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Missile-platform boundary. **Named evidence/example:** A missile is the guided weapon, while an aircraft, ship, submarine, vehicle or launcher is the delivery platform; one missile may have multiple launch configurations without becoming the platform itself. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **answer** requires a direct position on “Differentiate ballistic, cruise and guided rocket-artillery systems. Answer in about 150…”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Ballistic-cruise boundary. **Named evidence/example:** A ballistic missile receives powered boost and then follows a largely ballistic trajectory, whereas a cruise missile sustains guided atmospheric flight using an air-breathing engine; range alone does not define either class. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Hypersonic-class boundary. **Named evidence/example:** Hypersonic describes speed rather than one missile class: a boost-glide vehicle is rocket-boosted and manoeuvres while gliding, while a hypersonic cruise system uses sustained air-breathing propulsion such as a scramjet. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Rocket-artillery boundary. **Named evidence/example:** Pinaka is a multi-barrel rocket and guided-rocket artillery family for land fires; sharing rocket propulsion does not turn it into a strategic ballistic missile, cruise missile or space launch vehicle. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Missile-platform boundary. **Named evidence/example:** A missile is the guided weapon, while an aircraft, ship, submarine, vehicle or launcher is the delivery platform; one missile may have multiple launch configurations without becoming the platform itself. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** A ballistic missile receives powered boost and then follows a largely ballistic trajectory, whereas a cruise missile sustains guided atmospheric flight using an air-breathing engine; range alone does not define either class. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Hypersonic describes speed rather than one missile class: a boost-glide vehicle is rocket-boosted and manoeuvres while gliding, while a hypersonic cruise system uses sustained air-breathing propulsion such as a scramjet. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** Pinaka is a multi-barrel rocket and guided-rocket artillery family for land fires; sharing rocket propulsion does not turn it into a strategic ballistic missile, cruise missile or space launch vehicle. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** A missile is the guided weapon, while an aircraft, ship, submarine, vehicle or launcher is the delivery platform; one missile may have multiple launch configurations without becoming the platform itself. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Ballistic-cruise boundary. **Named evidence/example:** A ballistic missile receives powered boost and then follows a largely ballistic trajectory, whereas a cruise missile sustains guided atmospheric flight using an air-breathing engine; range alone does not define either class. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Hypersonic-class boundary. **Named evidence/example:** Hypersonic describes speed rather than one missile class: a boost-glide vehicle is rocket-boosted and manoeuvres while gliding, while a hypersonic cruise system uses sustained air-breathing propulsion such as a scramjet. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Rocket-artillery boundary. **Named evidence/example:** Pinaka is a multi-barrel rocket and guided-rocket artillery family for land fires; sharing rocket propulsion does not turn it into a strategic ballistic missile, cruise missile or space launch vehicle. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Missile-platform boundary. **Named evidence/example:** A missile is the guided weapon, while an aircraft, ship, submarine, vehicle or launcher is the delivery platform; one missile may have multiple launch configurations without becoming the platform itself. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “Differentiate ballistic, cruise and guided rocket-artillery systems. Answer in about 150…”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 3 - 15 MARKS

Question: Explain how subsystem integration and user evaluation convert research into a missile system. Answer in about 250 words.

Model thesis: Claim: MSS-cluster boundary. **Named evidence/example:** The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Subsystem-chain boundary. **Named evidence/example:** A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an integrated operational weapon. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Test-status ladder. **Named evidence/example:** Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Production-agency boundary. **Named evidence/example:** DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres.
- A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an integrated operational weapon.
- Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested.
- DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system.

Qualified conclusion: Claim: MSS-cluster boundary. **Named evidence/example:** The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Subsystem-chain boundary. **Named evidence/example:** A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an integrated operational weapon. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Test-status ladder. **Named evidence/example:** Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Production-agency boundary. **Named evidence/example:** DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **explain** requires a direct position on “Explain how subsystem integration and user evaluation convert research into a missile system.…”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: MSS-cluster boundary. **Named evidence/example:** The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Subsystem-chain boundary. **Named evidence/example:** A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an integrated operational weapon. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Test-status ladder. **Named evidence/example:** Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Production-agency boundary. **Named evidence/example:** DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an

integrated operational weapon. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

3. **Claim and named evidence:** Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: MSS-cluster boundary. **Named evidence/example:** The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Subsystem-chain boundary. **Named evidence/example:** A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an integrated operational weapon. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Test-status ladder. **Named evidence/example:** Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Production-agency boundary. **Named evidence/example:** DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “Explain how subsystem integration and user evaluation convert research into a missile system.…”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 4 - 15 MARKS

Question: Analyse India's missile ecosystem through role-based families rather than range lists. Answer in about 250 words.

Model thesis: Claim: Ballistic-cruise boundary. **Named evidence/example:** A ballistic missile receives powered boost and then follows a largely ballistic trajectory, whereas a cruise missile sustains guided atmospheric flight using an air-breathing engine; range alone does not define either class. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** SAM-AAM boundary. **Named evidence/example:** A surface-to-air missile is launched from the surface against aerial threats, while an air-to-air missile is launched from an aircraft; Akash and Astra therefore belong to different operational and launch-platform categories. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** ATGM boundary. **Named evidence/example:** An anti-tank guided missile is designed for armoured targets and uses a guidance and seeker logic distinct from strategic ballistic, cruise and air-defence systems; the Nag family belongs to this role. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Rocket-artillery boundary. **Named evidence/example:** Pinaka is a multi-barrel rocket and guided-rocket artillery family for land fires; sharing rocket propulsion does not turn it into a strategic ballistic missile, cruise missile or space launch vehicle. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** IGMDP-history boundary. **Named evidence/example:** The Integrated Guided Missile Development Programme ran from 1983 to March 2012 and covered Prithvi, Trishul, Akash, Nag and the Agni technology demonstrator; later families are not automatically IGMDP products. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** BrahMos-JV boundary. **Named evidence/example:** BrahMos is a supersonic cruise-missile system developed and produced through BrahMos Aerospace, the DRDO-NPOM joint venture; it is neither a purely solo domestic programme nor a ballistic missile. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- A ballistic missile receives powered boost and then follows a largely ballistic trajectory, whereas a cruise missile sustains guided atmospheric flight using an air-breathing engine; range alone does not define either class.
- A surface-to-air missile is launched from the surface against aerial threats, while an air-to-air missile is launched from an aircraft; Akash and Astra therefore belong to different operational and launch-platform categories.
- An anti-tank guided missile is designed for armoured targets and uses a guidance and seeker logic distinct from strategic ballistic, cruise and air-defence systems; the Nag family belongs to this role.
- Pinaka is a multi-barrel rocket and guided-rocket artillery family for land fires; sharing rocket propulsion does not turn it into a strategic ballistic missile, cruise missile or space launch vehicle.
- The Integrated Guided Missile Development Programme ran from 1983 to March 2012 and covered Prithvi, Trishul, Akash, Nag and the Agni technology demonstrator; later families are not automatically IGMDP products.
- BrahMos is a supersonic cruise-missile system developed and produced through BrahMos Aerospace, the DRDO-NPOM joint venture; it is neither a purely solo domestic programme nor a ballistic missile.

Qualified conclusion: Claim: Ballistic-cruise boundary. **Named evidence/example:** A ballistic missile receives powered boost and then follows a largely ballistic trajectory, whereas a cruise missile sustains guided atmospheric flight using an air-breathing engine; range alone does not define either class. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** SAM-AAM boundary. **Named evidence/example:** A surface-to-air missile is launched from the surface against aerial threats, while an air-to-air missile is launched from an aircraft; Akash and Astra therefore belong to different operational and launch-platform categories. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** ATGM boundary. **Named evidence/example:** An anti-tank guided missile is designed for armoured targets and uses a guidance and seeker logic distinct from strategic ballistic, cruise and air-defence systems; the Nag family belongs to this role. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Rocket-artillery boundary. **Named evidence/example:** Pinaka is a multi-barrel rocket and guided-rocket artillery family for land fires; sharing rocket propulsion does not turn it into a strategic ballistic missile, cruise missile or space launch vehicle. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** IGMDP-history boundary. **Named evidence/example:** The Integrated Guided Missile Development Programme ran from 1983 to March 2012 and covered Prithvi, Trishul, Akash, Nag and the Agni technology demonstrator; later families are not automatically IGMDP products. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** BrahMos-JV boundary. **Named evidence/example:** BrahMos is a supersonic cruise-missile system developed and produced through BrahMos Aerospace, the DRDO-NPOM joint venture; it is neither a purely solo domestic programme nor a ballistic missile. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **analyse** requires a direct position on “Analyse India's missile ecosystem through role-based families rather than range lists. Answer…”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: **Claim:** Ballistic-cruise boundary. **Named evidence/example:** A ballistic missile receives powered boost and then follows a largely ballistic trajectory, whereas a cruise missile sustains guided atmospheric flight using an air-breathing engine; range alone does not define either class. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** SAM-AAM boundary. **Named evidence/example:** A surface-to-air missile is launched from the surface against aerial threats, while an air-to-air missile is launched from an aircraft; Akash and Astra therefore belong to different operational and launch-platform categories. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** ATGM boundary. **Named evidence/example:** An anti-tank guided missile is designed for armoured targets and uses a guidance and seeker logic distinct from strategic ballistic, cruise and air-defence systems; the Nag family belongs to this role. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Rocket-artillery boundary. **Named evidence/example:** Pinaka is a multi-barrel rocket and guided-rocket artillery family for land fires; sharing rocket propulsion does not turn it into a strategic ballistic missile, cruise missile or space launch vehicle. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** IGMDP-history boundary. **Named evidence/example:** The Integrated Guided Missile Development Programme ran from 1983 to March 2012 and covered Prithvi, Trishul, Akash, Nag and the Agni technology demonstrator; later families are not automatically IGMDP products. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** BrahMos-JV boundary. **Named evidence/example:** BrahMos is a supersonic cruise-missile system developed and produced through BrahMos Aerospace, the DRDO-NPOM joint venture; it is neither a purely solo domestic programme nor a ballistic missile. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** A ballistic missile receives powered boost and then follows a largely ballistic trajectory, whereas a cruise missile sustains guided atmospheric flight using an air-breathing engine; range alone does not define either class. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** A surface-to-air missile is launched from the surface against aerial threats, while an air-to-air missile is launched from an aircraft; Akash and Astra therefore belong to different operational and launch-platform categories. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** An anti-tank guided missile is designed for armoured targets and uses a guidance and seeker logic distinct from strategic ballistic, cruise and air-defence systems; the Nag family belongs to this role. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** Pinaka is a multi-barrel rocket and guided-rocket artillery family for land fires; sharing rocket propulsion does not turn it into a strategic ballistic missile, cruise missile or space launch vehicle. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
5. **Claim and named evidence:** The Integrated Guided Missile Development Programme ran from 1983 to March 2012 and covered Prithvi, Trishul, Akash, Nag and the Agni technology demonstrator; later families are not automatically IGMDP products. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
6. **Claim and named evidence:** BrahMos is a supersonic cruise-missile system developed and produced through BrahMos Aerospace, the DRDO-NPOM joint venture; it is neither a purely solo domestic programme nor a ballistic missile. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Ballistic-cruise boundary. **Named evidence/example:** A ballistic missile receives powered boost and then follows a largely ballistic trajectory, whereas a cruise missile sustains guided atmospheric flight using an air-breathing engine; range alone does not define either class. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** SAM-AAM boundary. **Named evidence/example:** A surface-to-air missile is launched from the surface against aerial threats, while an air-to-air missile is launched from an aircraft; Akash and Astra therefore belong to different operational and launch-platform categories. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** ATGM boundary. **Named evidence/example:** An anti-tank guided missile is designed for armoured targets and uses a guidance and seeker logic distinct from strategic ballistic, cruise and air-defence systems; the Nag family belongs to this role. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Rocket-artillery boundary. **Named evidence/example:** Pinaka is a multi-barrel rocket and guided-rocket artillery family for land fires; sharing rocket propulsion does not turn it into a strategic ballistic missile, cruise missile or space launch vehicle. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** IGMDP-history boundary. **Named evidence/example:** The Integrated Guided Missile Development Programme ran from 1983 to March 2012 and covered Prithvi, Trishul, Akash, Nag and the Agni technology demonstrator; later families are not automatically IGMDP products. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** BrahMos-JV boundary. **Named evidence/example:** BrahMos is a supersonic cruise-missile system developed and produced through BrahMos Aerospace, the DRDO-NPOM joint venture; it is neither a purely solo domestic programme nor a ballistic missile. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “Analyse India's missile ecosystem through role-based families rather than range lists. Answer…”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 5 - 20 MARKS

Question: Evaluate the development-to-deployment chain for indigenous missile and air-defence capability. Answer in about 300 words.

Model thesis: Claim: DRDO-developer boundary. **Named evidence/example:** DRDO is the Ministry of Defence research-and-development organisation: it designs, develops, integrates and tests systems and facilitates production and induction; the armed services are users and production agencies manufacture at scale. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Requirement-user boundary. **Named evidence/example:** The Army, Navy and Air Force articulate operational requirements, participate in user evaluation and decide service acceptance; a DRDO developmental result cannot be rewritten as service induction. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Subsystem-chain boundary. **Named evidence/example:** A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an integrated operational weapon. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Test-status ladder. **Named evidence/example:**

Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** AoN-contract boundary. **Named evidence/example:** Defence Acquisition Council Acceptance of Necessity is in-principle approval of a requirement and category, not a signed contract, delivery, induction or operational deployment. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Production-agency boundary. **Named evidence/example:** DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Integrated-air-defence boundary. **Named evidence/example:** Air and missile defence is a sensor, command-and-control and interceptor architecture; an announced integrated mission or one interceptor does not prove a nationwide deployed shield. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Volatile-capability boundary. **Named evidence/example:** Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- DRDO is the Ministry of Defence research-and-development organisation: it designs, develops, integrates and tests systems and facilitates production and induction; the armed services are users and production agencies manufacture at scale.
- The Army, Navy and Air Force articulate operational requirements, participate in user evaluation and decide service acceptance; a DRDO developmental result cannot be rewritten as service induction.
- A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an integrated operational weapon.
- Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested.
- Defence Acquisition Council Acceptance of Necessity is in-principle approval of a requirement and category, not a signed contract, delivery, induction or operational deployment.
- DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system.
- Air and missile defence is a sensor, command-and-control and interceptor architecture; an announced integrated mission or one interceptor does not prove a nationwide deployed shield.
- Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers.

Qualified conclusion: Claim: DRDO-developer boundary. **Named evidence/example:** DRDO is the Ministry of Defence research-and-development organisation: it designs, develops, integrates and tests systems and facilitates production and induction; the armed services are users and production agencies manufacture at scale. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Requirement-user boundary. **Named evidence/example:** The Army, Navy and Air Force articulate operational requirements, participate in user evaluation and decide service acceptance; a DRDO developmental result cannot be rewritten as service induction. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Subsystem-chain boundary. **Named evidence/example:** A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an integrated operational weapon. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Test-status ladder. **Named evidence/example:** Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** AoN-contract boundary. **Named evidence/example:** Defence Acquisition Council Acceptance of Necessity is in-principle approval of a requirement and category, not a signed contract, delivery, induction or operational deployment. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Production-agency boundary. **Named evidence/example:** DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Integrated-air-defence boundary. **Named evidence/example:** Air and missile defence is a sensor, command-and-control and interceptor architecture; an announced integrated mission or one interceptor does not prove a nationwide deployed shield. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Volatile-capability boundary. **Named evidence/example:** Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **evaluate** requires a direct position on “Evaluate the development-to-deployment chain for indigenous missile and air-defence…”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: DRDO-developer boundary. **Named evidence/example:** DRDO is the Ministry of Defence research-and-development organisation: it designs, develops, integrates and tests systems and facilitates production and induction; the armed services are users and production agencies manufacture at scale. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Requirement-user boundary. **Named evidence/example:** The Army, Navy and Air Force articulate operational requirements, participate in user evaluation and decide service acceptance; a DRDO developmental result cannot be rewritten as service induction. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Subsystem-chain boundary. **Named evidence/example:** A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an integrated operational weapon. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Test-status ladder. **Named evidence/example:** Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** AoN-contract boundary. **Named evidence/example:** Defence Acquisition Council Acceptance of Necessity is in-principle approval of a requirement and category, not a signed contract, delivery, induction or operational deployment. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Production-agency boundary. **Named evidence/example:** DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Integrated-air-defence boundary. **Named evidence/example:** Air and missile defence is a sensor, command-and-control and interceptor architecture; an announced integrated mission or one interceptor does not prove a nationwide deployed shield. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Volatile-capability boundary. **Named evidence/example:** Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** DRDO is the Ministry of Defence research-and-development organisation: it designs, develops, integrates and tests systems and facilitates production and induction; the armed services are users and production agencies manufacture at scale. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** The Army, Navy and Air Force articulate operational requirements, participate in user evaluation and decide service acceptance; a DRDO developmental result cannot be rewritten as service induction. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an integrated operational weapon. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
5. **Claim and named evidence:** Defence Acquisition Council Acceptance of Necessity is in-principle approval of a requirement and category, not a signed contract, delivery, induction or operational deployment. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

6. **Claim and named evidence:** DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
7. **Claim and named evidence:** Air and missile defence is a sensor, command-and-control and interceptor architecture; an announced integrated mission or one interceptor does not prove a nationwide deployed shield. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
8. **Claim and named evidence:** Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: DRDO-developer boundary. **Named evidence/example:** DRDO is the Ministry of Defence research-and-development organisation: it designs, develops, integrates and tests systems and facilitates production and induction; the armed services are users and production agencies manufacture at scale. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Requirement-user boundary. **Named evidence/example:** The Army, Navy and Air Force articulate operational requirements, participate in user evaluation and decide service acceptance; a DRDO developmental result cannot be rewritten as service induction. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Subsystem-chain boundary. **Named evidence/example:** A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an integrated operational weapon. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Test-status ladder. **Named evidence/example:** Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** AoN-contract boundary. **Named evidence/example:** Defence Acquisition Council Acceptance of Necessity is in-principle approval of a requirement and category, not a signed contract, delivery, induction or operational deployment. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Production-agency boundary. **Named evidence/example:** DRDO-developed systems require production partners such as DPSUs, joint ventures or private firms; developer identity does not establish who manufactures, maintains or upgrades the fielded system. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Integrated-air-defence boundary. **Named evidence/example:** Air and missile defence is a sensor, command-and-control and interceptor architecture; an announced integrated mission or one interceptor does not prove a nationwide deployed shield. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Volatile-capability boundary. **Named evidence/example:** Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 20 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “Evaluate the development-to-deployment chain for indigenous missile and air-defence…”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 6 - 20 MARKS

Question: Discuss strategic autonomy in defence R&D while preserving secrecy, status and institutional boundaries. Answer in about 300 words.

Model thesis: Claim: MSS-cluster boundary. **Named evidence/example:** The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Subsystem-chain boundary. **Named evidence/example:** A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an integrated operational weapon. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** BrahMos-JV boundary. **Named evidence/example:** BrahMos is a supersonic cruise-missile system developed and produced through BrahMos Aerospace, the DRDO-NPOM joint venture; it is neither a purely solo domestic programme nor a ballistic missile. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Test-status ladder. **Named evidence/example:** Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Integrated-air-defence boundary. **Named evidence/example:** Air and missile defence is a sensor, command-and-control and interceptor architecture; an announced integrated mission or one interceptor does not prove a nationwide deployed shield. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Stealth-detection boundary. **Named evidence/example:** Stealth seeks lower observability through shaping, materials and operating tactics, but it does not mean invisibility; detection depends on sensors, frequency, geometry, processing and the operational environment. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Volatile-capability boundary. **Named evidence/example:** Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres.
- A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an integrated operational weapon.
- BrahMos is a supersonic cruise-missile system developed and produced through BrahMos Aerospace, the DRDO-NPOM joint venture; it is neither a purely solo domestic programme nor a ballistic missile.
- Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested.
- Air and missile defence is a sensor, command-and-control and interceptor architecture; an announced integrated mission or one interceptor does not prove a nationwide deployed shield.
- Stealth seeks lower observability through shaping, materials and operating tactics, but it does not mean invisibility; detection depends on sensors, frequency, geometry, processing and the operational environment.
- Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers.

Qualified conclusion: Claim: MSS-cluster boundary. **Named evidence/example:** The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Subsystem-chain boundary. **Named evidence/example:** A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an integrated operational weapon. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** BrahMos-JV boundary. **Named evidence/example:** BrahMos is a supersonic cruise-missile system developed and produced through BrahMos Aerospace, the DRDO-NPOM joint venture; it is neither a purely solo domestic programme nor a ballistic missile. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Test-status ladder. **Named evidence/example:** Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Integrated-air-defence boundary. **Named evidence/example:** Air and missile defence is a sensor, command-and-control and interceptor architecture; an announced integrated mission or one interceptor does not prove a nationwide deployed shield. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Stealth-detection boundary. **Named evidence/example:** Stealth seeks lower observability through shaping, materials and operating tactics, but it does not mean invisibility; detection depends on sensors, frequency, geometry, processing and the operational environment. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Volatile-capability boundary. **Named evidence/example:** Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **discuss** requires a direct position on "Discuss strategic autonomy in defence R&D while preserving secrecy, status and institutional…", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: MSS-cluster boundary. **Named evidence/example:** The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Subsystem-chain boundary. **Named evidence/example:** A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an integrated operational weapon. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** BrahMos-JV boundary. **Named evidence/example:** BrahMos is a supersonic cruise-missile system developed and produced through BrahMos Aerospace, the DRDO-NPOM joint venture; it is neither a purely solo domestic programme nor a ballistic missile. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Test-status ladder. **Named evidence/example:** Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Integrated-air-defence boundary. **Named evidence/example:** Air and missile defence is a sensor, command-and-control and interceptor architecture; an announced integrated mission or one interceptor does not prove a nationwide deployed shield. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Stealth-detection boundary. **Named evidence/example:** Stealth seeks lower observability through shaping, materials and operating tactics, but it does not mean invisibility; detection depends on sensors, frequency, geometry, processing and the operational environment. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Volatile-capability boundary. **Named evidence/example:** Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

2. **Claim and named evidence:** A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an integrated operational weapon. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
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Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: MSS-cluster boundary. **Named evidence/example:** The Missiles and Strategic Systems cluster is responsible for missile and strategic-system design and development through DRDL, RCI, ASL, TBRL and ITR plus supporting test, integration and analysis centres. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Subsystem-chain boundary. **Named evidence/example:** A missile system combines airframe, propulsion, navigation, guidance, homing or seeker, warhead, launch system and command-and-control; success of one subsystem is not proof of an integrated operational weapon. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** BrahMos-JV boundary. **Named evidence/example:** BrahMos is a supersonic cruise-missile system developed and produced through BrahMos Aerospace, the DRDO-NPOM joint venture; it is neither a purely solo domestic programme nor a ballistic missile. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Test-status ladder. **Named evidence/example:** Developmental trial, user evaluation, ready-for-induction language, contract, induction and deployment are separate rungs; a successful test proves only the proposition and configuration officially tested. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Integrated-air-defence boundary. **Named evidence/example:** Air and missile defence is a sensor, command-and-control and interceptor architecture; an announced integrated mission or one interceptor does not prove a nationwide deployed shield. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Stealth-detection boundary. **Named evidence/example:** Stealth seeks lower observability through shaping, materials and operating tactics, but it does not mean invisibility; detection depends on sensors, frequency, geometry, processing and the operational environment. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Volatile-capability boundary. **Named evidence/example:** Missile range, speed, stages, guidance, warhead, payload, deployment, inventory and exact test outcome require a dated official source; qualitative role labels cannot be expanded into unaudited numbers. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 20 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “Discuss strategic autonomy in defence R&D while preserving secrecy, status and institutional…”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.